dark matter must lie in these far-out regions, well beyond the visible edge of each spiral galaxy. Thanks to Rubin's work, we now call these mysterious zones "dark matter haloes."

This halo problem exists under our noses, right in the Milky Way. From galaxy to galaxy and from cluster to cluster, the discrepancy between the mass tallied from visible objects and the objects' mass estimated from total gravity ranges from a factor of a few up to (in some cases) a factor of many hundreds. Across the universe, the discrepancy averages to a factor of six: cosmic dark matter has about six times the total gravity of all the visible matter.

Further research has revealed that the dark matter cannot consist of ordinary matter that happens to be under-luminous, or nonluminous. This conclusion rests on two lines of reasoning. First, we can eliminate with near-certainty all plausible familiar candidates, like the suspects in a police lineup. Could the dark matter reside in black holes? No, we think that we would have detected this many black holes from their gravitational effects on nearby stars. Could it be dark clouds? No, they would absorb or otherwise interact with light from stars behind them, which bona fide dark matter doesn't do. Could it be interstellar (or intergalactic) rogue planets, asteroids, and comets, all of which produce no light of their own? It's hard to believe that the universe would manufacture six times as much mass in planets as in stars. That would mean six thousand Jupiters for every star in the galaxy, or worse yet, two million Earths. In our own solar system, for example, everything that is not the Sun adds up to less than one fifth of one percent of the Sun's mass.

More direct evidence for the strange nature of dark matter comes from the relative amount of hydrogen and helium in the universe. Together, these numbers provide a cosmic fingerprint left behind by the early universe. To a close approximation, nuclear fusion during the first few minutes after the big bang left behind one helium nucleus for every ten hydrogen nuclei (which are, themselves, simply protons). Calculations show that if most of the dark matter had involved itself in nuclear fusion, there would be much more helium relative to hydrogen in the universe. From this we conclude that most of the dark matter—hence, most of the mass in the universe—does not participate in nuclear fusion, which disqualifies it as "ordinary" matter, whose essence lies in a willingness to participate in the atomic and nuclear forces that shape matter as we know it. Detailed observations of the cosmic microwave